AE4446: *Airport and Cargo Operations* - Gate Assignment

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https://blog.klm.com/nl/10-dingen-die-je-niet-wist-over-schiphol/

Outline

What is gate assignment?

A gate is an **area in an airport terminal that controls access to a passenger aircraft**. While the exact specifications vary from airport to airport and country to country, most gates consist of a seated waiting area, a counter, and a doorway leading to the aircraft

Every aircraft landing/taking-off from an airport must be assigned to a gate to disembark/embark passengers

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A review on airport gate assignment problems: Single versus multi objective approaches☆

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ABSTRACT

Assigning aircraft to gates is an important decision problem that airport professionals face every day. The solution of this problem has raised a significant research effort and many variants of this problem have been studied. In this paper, we review past work with a focus on identifying types of formulations, classifying objectives, and categorising solution methods. The review indicates that there is no standard formulation, that passenger oriented objectives are most common, and that more recent work are multi-objective. In terms of solution methods, heuristic and metaheuristic approaches are dominant which provides an opportunity to develop exact and approximate approaches both for the single and multi-objective problems.

The **Gate Assignment Problem (GAP)** is a very complicated one

- different airport sizes
- different airport roles (transfer passengers)
- different stakeholders

Different stakeholders

Three main stakeholders with (competing) interests in the GAP problem

- **airline operators**: easy access to terminals, short ground times
- **passengers**: quick and smooth boarding, short walking distances, easy access to airport amenities
- **airport operators**: increase in revenue, maximization of efficiency of airport resources, minimization of congestion, delays, etc.

It might result in a **tug of war game**

GAP Mathematical Modeling

There are several ways (methodologically speaking) to model the GAP

- **assignment problem** (as the name of the problem suggests)
- **multi-commodity flow problem**

and many different **practical considerations and features** that might complicate the model and make it more realistic

We will discuss a **basic (yet, non-trivial) implementation** and some possible improvements

GAP Mathematical Modeling: Sets

Two main **sets** involved

Credit: *nieuws.schiphol*

Credit: *next.atl*

Set of **gates** $\mathcal{G} = \{0, 1, \cdots, m\}$, indexed by g

Gate 0 is a fictitious gate, i.e., the **airport apron**, with infinite capacity

GAP Mathematical Modeling: Sets

Schiphol has gates divided into its own set of 8 piers **B, C, D, E, F, G, H, M**

Credit: *nieuws.schiphol*

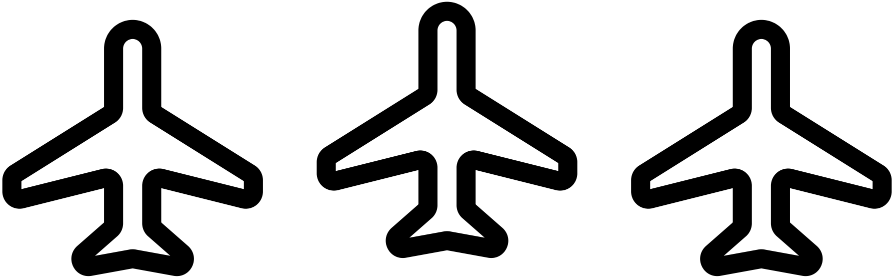
Why no pier **A**? The English-pronounced A is too similar to the Dutch-pronounced E. This would cause confusion in communication. Why a jump from **H** to **M**? **I** and **J** are visually too similar, **K** and **L** were reserved in the past for a potential raising of gates **F** and **G**.

GAP Mathematical Modeling: Sets

Credits: *googleMaps*

Some airports have **no physical gates (or jet bridges, more properly)** at all!

GAP Mathematical Modeling: Sets

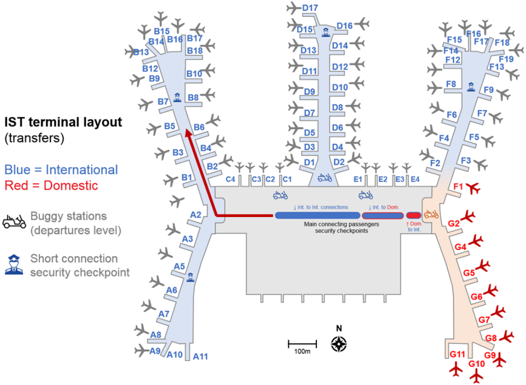


Credits: *iconoir*

Set of **aircraft** $\mathcal{A} = \{1, \cdots, n\}$, indexed by i or j

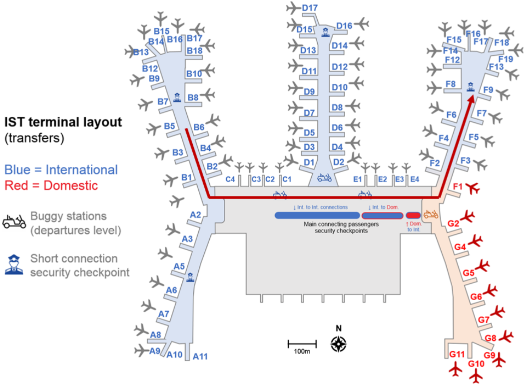
As an aircraft generally performs consecutive flights, we consider the **arrival time** of the first flight A_i and the **departure time** of the second flight D_i

GAP Mathematical Modeling: Parameters



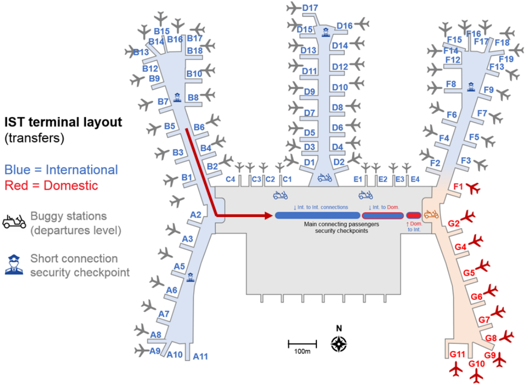
$D_{0,g}$: walking distance between check-in area (**0**) and gate g. Focus on **outbound passengers**

GAP Mathematical Modeling: Parameters



D_{g_1,g_2}: walking distance between gates g_1 and g_2. Focus on **transfer passengers**

GAP Mathematical Modeling: Parameters



$D_{g,0}$: walking distance between gate g and luggage area (**0**). Focus on **inbound passengers**

GAP Mathematical Modeling: Parameters

Information on passengers is crucial

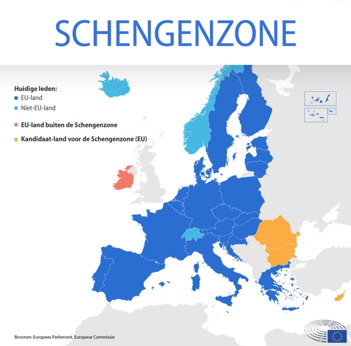
- $P_{i,j}$: number of passengers between aircraft i (inbound flight) and aircraft j (outbound flight) $\rightarrow \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{A}} P_{i,j}$ is overall amount of **transfer passengers**

- $P_{0,i}$: number of passengers from check-in to aircraft i (outbound flight) $\rightarrow \sum_{i \in \mathcal{A}} P_{0,i}$ is overall amount of **outbound passengers**

- $P_{i,0}$: number of passengers from aircraft i (inbound flight) to luggage area $\rightarrow \sum_{i \in \mathcal{A}} P_{i,0}$ is overall amount of **inbound passengers**

Different airports have a **different distribution** across the three types of passengers

GAP Mathematical Modeling: Parameters



Credits: *https://www.europarl.europa.eu*

Not all gates might be available for a specific aircraft: $\mathcal{G}_i$: subset of gates that can be used by aircraft i, $\mathcal{A}_g$: subset of aircraft that can use gate g

GAP Mathematical Modeling: Parameters

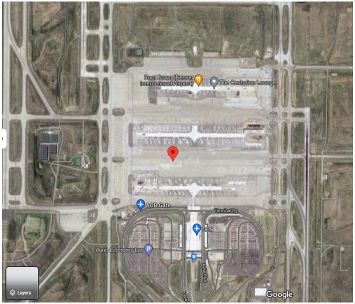
Preferences of airlines are also important and/or constraining. At **AMS**, Easyjet only uses gates **M**

Credits: *https://vanhiertottimboektoe.nl/*

If we consider $\mathcal{A}_{U2} \subseteq \mathcal{A}$ as the subset of Easyjet aircraft being handled by **AMS** in a given day and $\mathcal{G}_M \subseteq \mathcal{G}$ as the subset of M gates, then $\forall a \in \mathcal{A}_{U2}$ **only 8 possible assignments are possible (Schiphol has 8 M gates)**

GAP Mathematical Modeling: Parameters

In a similar fashion, at **DEN** SouthWest only uses Concourse 3



Credit: *googleMaps*

If we consider $\mathcal{A}_{WN} \subseteq \mathcal{A}$ as the subset of Southwest aircraft being handled by **DEN** in a given day and $\mathcal{G}_3 \subseteq \mathcal{G}$ as the subset of the gates belonging to concourse 3, then $\forall a \in \mathcal{A}_{WN}$ **"only" 47 possible assignments are possible (concourse 3 has 47 gates)**

GAP Mathematical Modeling: Decision Variables

In principle, only one set of decision variables is needed

$$x_{i,g} = \begin{cases} 1 & \text{if aircraft } i \text{ is assigned to gate } g \\ 0 & \text{otherwise} \end{cases}$$

Additional decision variables are needed for more complicated versions of the GAP

GAP Mathematical Modeling: Basic Constraints

Our goal is to **assign every aircraft to a gate**. In this initial and basic version of the GAP, it can be either a physical gate (with a capacity of 1 aircraft), or the apron (∞ capacity, but comes with other issues)

$$\sum_{g \in \mathcal{G}_i} x_{i,g} = 1 \quad \forall i \in \mathcal{A}$$

In addition, we want to make sure that **aircraft with overlapping schedules cannot be assigned to the same gate (considering all the common gates between the two aircraft)**

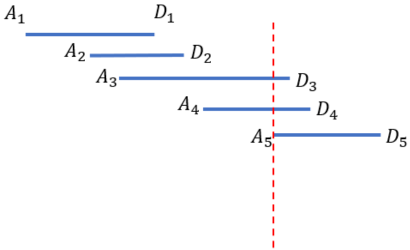
$$x_{i,g} x_{j,g} (D_j - A_i)(D_i - A_j) \leq 0 \quad \forall i \in \mathcal{A}, j \in \mathcal{A}, g \in \mathcal{G}_i \cap \mathcal{G}_j$$

We can only assign two aircraft to the same gate if their schedules do not overlap $\rightarrow$ $A_i \geq D_j \vee A_j \geq D_i$

GAP Mathematical Modeling: Basic Constraints

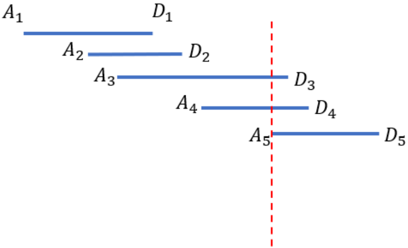
The previous constraint is non-linear, but such constraint is not needed if we map $\forall i \in \mathcal{A}$ the subset of other aircraft $\mathcal{A}^i_{inc}$ whose schedules overlap with i

We can sort $\mathcal{A}$ by increasing values of arrival A_i, so that if $j > i \rightarrow A_j \geq A_i$



$$\mathcal{A}^i_{inc} = \{j \mid D_j \geq A_i \;\; \forall j < i\}$$

GAP Mathematical Modeling: Basic Constraints



$$\mathcal{A}^1_{inc} = \{\}$$
$$\mathcal{A}^2_{inc} = \{1\}$$
$$\mathcal{A}^3_{inc} = \{1,2\}$$
$$\mathcal{A}^4_{inc} = \{3\}$$
$$\mathcal{A}^5_{inc} = \{3,4\}$$

Note: $\mathcal{A}^1_{inc} = \{\}$ does not mean aircraft 1 is compatible with all the other aircraft. In fact, it is labeled as incompatible with both aircraft 2 and 3. In this example, all incompatible pairs are **(1,2), (1,3), (2,3), (3,4), (3,5), (4,5)**. We can impose each constraint mapping an incompatibility as

$$x_{j,g} + x_{i,g} \leq 1 \;\; \forall i \in \mathcal{A}, j \in \mathcal{A}^i_{inc}, g \in \mathcal{G}_i \cap \mathcal{G}_j$$

$\mathcal{G}_i \cap \mathcal{G}_j$ **ensures that we only add constraints for those gates that time-incompatible aircraft i and j can both use!**

GAP Mathematical Modeling: Objective

If we focus on passengers, one option is to **minimize the overall traveled distance** (or time, assuming constant speed)

$$
\begin{aligned}
\min \quad & \sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_i} P_{0,i} D_{0,g} x_{i,g} + && \textbf{(outbound passengers)} \\
& \sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_i} P_{i,0} D_{g,0} x_{i,g} + && \textbf{(inbound passengers)} \\
& \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{A} \setminus \{i\}} \sum_{g_1 \in \mathcal{G}_i} \sum_{g_2 \in \mathcal{G}_j} P_{i,j} D_{g_1,g_2} x_{i,g_1} x_{j,g_2} && \textbf{(transfer passengers)}
\end{aligned}
$$

$x_{i,g_1} x_{j,g_2}$ is a **non-linear** term

GAP Mathematical Modeling: Objective

It can be **linearized** by introducing an additional set of decision variables

$$y_{i,g_1,j,g_2} = \begin{cases} 1, & \text{if aircraft } i \text{ is assigned to gate } g_1 \\ & \text{and aircraft } j \text{ to gate } g_2 \\ 0, & \text{otherwise} \end{cases}$$

and the additional sets of constraints

$$\begin{aligned} y_{i,g_1,j,g_2} &\geq x_{i,g_1} + x_{j,g_2} - 1 && \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \\ y_{i,g_1,j,g_2} &\leq x_{i,g_1} && \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \\ y_{i,g_1,j,g_2} &\leq x_{j,g_2} && \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \end{aligned}$$

GAP Mathematical Modeling: Objective

$$y_{i,g_1,j,g_2} \geq x_{i,g_1} + x_{j,g_2} - 1 \qquad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j$$
$$y_{i,g_1,j,g_2} \leq x_{i,g_1} \qquad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j$$
$$y_{i,g_1,j,g_2} \leq x_{j,g_2} \qquad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j$$

y_{i,g_1,j,g_2} **forced to be unitary** if and only if both x_{i,g_1} and x_{j,g_2} are unitary. From first constraint set: $1 \geq 1 + 1 - 1 = 1$

y_{i,g_1,j,g_2} **forced to be zero** as soon as one between x_{i,g_1} and x_{j,g_2} is zero. From second or third constraint set: $0 \leq 0$

Note: we define y decision variables $\forall i \in \mathcal{A}, j \in \mathcal{A} : j > i$ to limit ourselves to the **minimum number of decision variables necessary and avoid redundancy**. For example, assigning aircraft 1 to gate 7 and aircraft 2 to gate 3 ($y_{1,7,2,3}$: **necessary**) is the same as assigning aircraft 2 to gate 3 and aircraft 1 to gate 7 ($y_{2,3,1,7}$: **redundant**)

GAP Mathematical Modeling: Objective

Linearized objective function

$$\min \quad \sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_i} P_{0,i} D_{0,g} x_{i,g} + \quad \textbf{(outbound passengers)}$$

$$\sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_i} P_{i,0} D_{g,0} x_{i,g} + \quad \textbf{(inbound passengers)}$$

$$\sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{A}: j > i} \sum_{g_1 \in \mathcal{G}_i} \sum_{g_2 \in \mathcal{G}_j} (P_{i,j} + P_{j,i}) D_{g_1,g_2} y_{i,g_1,j,g_2}$$

$$\textbf{(transfer passengers)}$$

Note: because of the way we defined y variables, now in the **transfer passenger** term we need to capture both $P_{i,j}$ and $P_{j,i}$ because we have a single combined y decision variable mapping the passenger flow from i to j and vice-versa

GAP Mathematical Modeling: basic model

$$
\begin{aligned}
\min \quad & \sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_i} P_{0,i} D_{0,g} x_{i,g} + \quad \textbf{(outbound passengers)} \\
& \sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_i} P_{i,0} D_{g,0} x_{i,g} + \quad \textbf{(inbound passengers)} \\
& \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{A}: j > i} \sum_{g_1 \in \mathcal{G}_i} \sum_{g_2 \in \mathcal{G}_j} (P_{i,j} + P_{j,i}) D_{g_1,g_2} y_{i,g_1,j,g_2} \quad \textbf{(transfer passengers)}
\end{aligned}
\tag{1}
$$

subject to:

$$\sum_{g \in \mathcal{G}_i} x_{i,g} = 1 \qquad \forall i \in \mathcal{A} \tag{2}$$

$$x_{j,g} + x_{i,g} \leq 1 \qquad \forall i \in \mathcal{A}, j \in \mathcal{A}^i_{inc}, g \in \mathcal{G}_i \cap \mathcal{G}_j \tag{3}$$

GAP Mathematical Modeling: basic model (continued)

$$y_{i,g_1,j,g_2} \geq x_{i,g_1} + x_{j,g_2} - 1 \quad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \tag{4}$$

$$y_{i,g_1,j,g_2} \leq x_{i,g_1} \quad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \tag{5}$$

$$y_{i,g_1,j,g_2} \leq x_{j,g_2} \quad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \tag{6}$$

$$x_{i,g} \in \{0,1\} \quad \forall i \in \mathcal{A}, g \in \mathcal{G}_i \tag{7}$$

$$y_{i,g_1,j,g_2} \in \{0,1\} \quad \forall i \in \mathcal{A}, j \in \mathcal{A} : j > i, g_1 \in \mathcal{G}_i, g_2 \in \mathcal{G}_j \tag{8}$$

Constraint 2 ensures that each aircraft is assigned to exactly one gate. Constraint 3 ensures that incompatible aircraft are assigned to different gates. Constraints 4-6 are linearization constraints. Constraints 7-8 define the nature of the decision variables

GAP Mathematical Modeling: Objective

Passenger-centric objectives

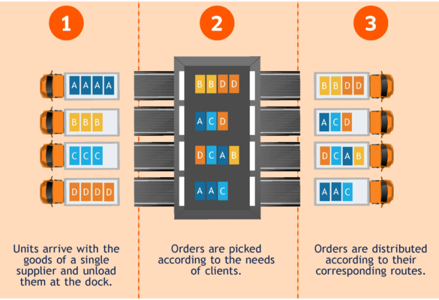
Table 2
Passenger oriented objective functions.

Objective	References	Number of objectives single (S)/Multiple (M)
Minimize total walking distance of all passengers or transfer passengers	Braksma and Shortreed (1971)[20]	S
	Mangoubi and Mathaisel (1985)[8]	S
	Bihr (1990)[21]	S
	Cheng (1997)[22]	M
	Haghani and Chen (1998)[23]	S
	Xu and Bailey (2001)[24]	S
	Yan and Huo (2001)[13]	M
	Ding et al. (2005)[25]	M
	Lim et al. (2005)[19]	M
	Hu and Paolo (2009)[46]	M
	Drexl and Nikulin (2008)[27]	M
	Pintea et al. (2008)[28]	M
	Cheng et al. (2012)[29]	S
	Yu et al. (2016, 2017)[18,41]	M
	Deng et al. (2017)[30]	M
	Dell'orco et al. (2017)[31]	M
	Daş (2017)[32]	M
	Mokhtarimousavi et al. (2018)[33]	M
	Dijk et al. (2018)[49]	S
Minimize average passenger walking distance	Babic et al. (1984)[34]	S
Minimize expected passenger discomfort	Maharjan and Matis (2012)[15]	M
Passenger waiting time/Transit time		
Minimize total passenger waiting time	Yan and Huo (2001)[13]	M
	Yan and Tang (2007)[14]	S
Minimize passenger transit time	Kim et al. (2013, 2017)[35,51]	M
Minimize estimated transfer time for passengers	Benlic et al. (2017)[36]	M
Baggage distance		
Minimize baggage transferring distance	Cheng (1997)[22]	M
	Hu and Paolo (2009)[46]	M

- total traveled distance
- average traveled distance
- total traveled time $\rightarrow$ **role of apron is key (it generally increases traveled time quite considerably)**

GAP Mathematical Modeling: Objective

Handling transfer passengers is similar in nature to **Cross-docking**: a practice in logistics of unloading materials from a manufacturer or mode of transportation directly to the customer or another mode of transportation, with little or no storage in between (Wikipedia)



Credits: *waredock*

GAP Mathematical Modeling: Objective

So far we have not considered any cost that the airline or the airport might (and will) incur as a consequence of the gate assignment

If we focus on airport or airline objectives, there is even more **variability**

Assigning aircraft to the apron is **not efficient**. Transfer from and to the aircraft stationed at the apron by buses $\rightarrow$ **increase in passenger waiting time** and **decrease in satisfaction especially in extreme cold or hot weather**

$$\min \quad \sum_{i \in \mathcal{A}} x_{i,0} \quad \textbf{(un-gated aircraft)}$$

An even better methodological (and practical) approach is to consider **the actual number of remote gates** (hence, the apron does not have ∞ capacity any longer)

GAP Mathematical Modeling: Objective

Considering **RTM**, the apron is divided into 12 remote stands

We could divide the overall set of gates $\mathcal{G}$ into a set of remote stands $\mathcal{G}_{rs} = \{1, \cdots, N_{rs}\}$ and a set of actual gates $\mathcal{G}_g = \{N_{rs}+1, \cdots, N_{rs}+N_g\}$ (if any)

$$\min \quad \sum_{i \in \mathcal{A}} \sum_{g \in \mathcal{G}_{rs}} x_{i,g} \quad \textbf{(un-gated aircraft)}$$

GAP Mathematical Modeling: Objective

Sometimes an aircraft must **wait hours before consecutive flights**, or **should be moved to a different gate to improve turnaround**

In this case, our assumption that an aircraft stays in the same position from A_i to D_i should be relaxed, as the two gates might be different or the aircraft might be moved somewhere else in the meantime

The approach considering the set of aircraft $i \in \mathcal{A}$ cannot be used. Switch to flights or, even better, activities: **arrivals (flight), departures (flight), parking**

GAP Mathematical Modeling: Objective

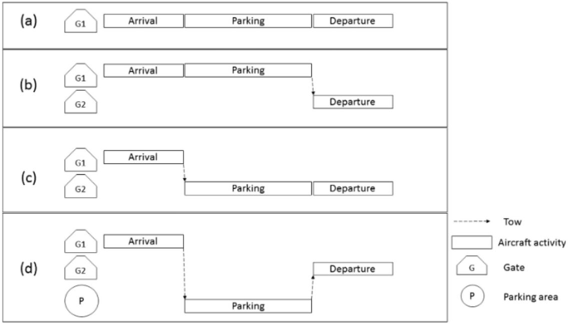


Fig. 7. Possible assignments for an aircraft at an airport.

Credits: *MIP-based heuristics for solving robust gate assignment problems*

With such approach, **minimization of towing operations** can be implemented, as **they are generally quite expensive**

GAP Mathematical Modeling: Objective

Airport/Airline objectives

Table 3
Airline/Airport oriented objective functions.

Objective	References	Number of objectives Single (S)/Multiple (M)
Objectives related to un-gated flights		
Minimize number of un-gated flights	Ding et al.(2005)[25]	M
	Dorndorf et al. (2008, 2012, 2017)[39,40,50]	M
	Drexl and Nikulin (2008)[27]	M
	Pintea et al. (2008)[28]	M
	Neuman and Atkin (2013)[11]	M
	Deng et al. (2017)[30]	M
	Kaliszewski et al.(2017)[47]	M
Minimize number of flights assigned to remote gates	Cheng (1997)[22]	M
	Tang and Wang (2013)[16]	M
	Dell'orco et al. (2017)[31]	M
Maximize total duration of flights assigned to gates	Genç et al. (2012)[74]	S
Minimize cost of not assigning a flight to a gate	Kumar and Beirlaire (2014)[10]	M
Objectives related to towing operations		
Minimize number of towing moves	Dorndorf et al. (2007b, 2008, 2012, 2017)[9,39,40,50]	M
	Nikulin and Drexl (2010)[12]	M
	Benlic et al. (2017)[36]	M
	Dijk et al. (2018)[49]	S
Minimize towing cost	Kumar and Beirlaire (2014)[10]	M
	Yu et al. (2016, 2017)[18,41]	M
	Mokhtarimousavi et al. (2018)[33]	M
Maximize number of arriving flights and the subsequent departing flights assigned to the same gate, if served by the same aircraft	Tang and Wang (2013)[16]	M
Objectives related to taxiing operations		
Minimize fuel consumption of taxiing operations	Maharjan and Matis (2012)[15]	M
Minimize taxi time	Kim et al. (2013, 2017)[35, 51]	M
	Ding and Zhang (2013)[80]	M
Objectives related to gate/ delay costs		
Minimize aircraft waiting time for a gate (waiting delay)	Cheng (1997)[22]	M
	Hu and Paolo (2009)[46]	M
Minimize total delay penalties	Lim et al. (2005)[19]	M
Minimize sum of waiting times for an aircraft for a gate	Kaliszewski et al.(2017)[47]	M
Minimize cost of assigning an aircraft to a gate	van Schaijk and Visser (2017)[52]	S
Objectives related to gates/ airline preferences		
Maximize total flight-gate preferences	Dorndorf et al. (2007b,2008,2012,2017)[9,39,40,50]	M
	Drexl and Nikulin (2008)[27]	M
	Nikulin and Drexl (2010)[12]	M
	Jaehn (2010)[42]	S
	Kumar and Beirlaire (2014)[10]	M
	Neuman and Atkin (2013)[11]	M
	Benlic et al. (2017)[36]	M
Maximize aircraft-gate size compatability	Neuman and Atkin (2013)[11]	M
	Benlic et al. (2017)[36]	M
Objectives related to airline/ airport specific goals		
Maximize number of departing flights assigned to short distances to target airline VIP lounges	Tang and Wang (2013)[16]	M
Maximize number of flights assigned to gates with short distance to customs		
Maximize number of passengers at gates close to shopping facilities to increase potential revenues	Daş (2017)[32]	M
Maximize potential airport commercial revenues	Dijk et al. (2018)[49]	S
Maximize number of passengers at gates	Daş (2017)[32]	M
	Dijk et al.(2018)[49]	S

- maximize gated aircraft
- minimize towing operations
- maximize total commercial revenue

Gates vs Stands

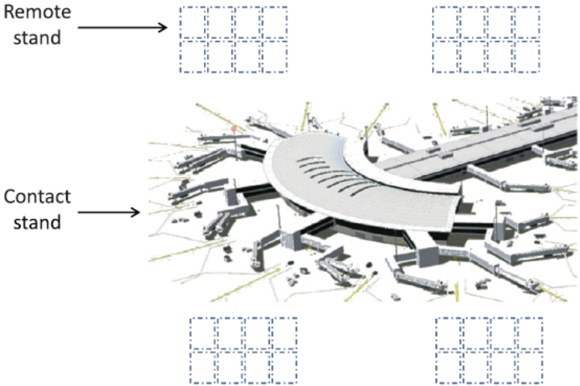
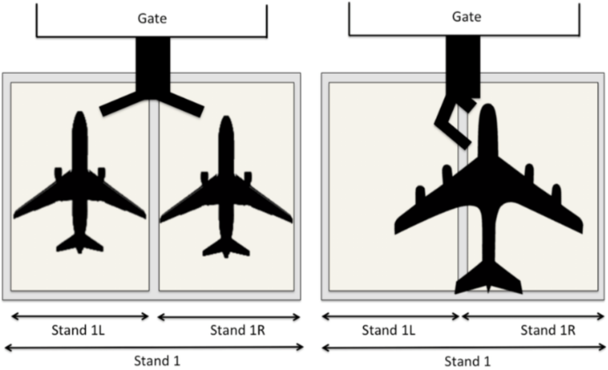


Fig. 1. Airport stands.

Credits: *Exact and heuristic approaches to the airport stand allocation problem*

Gates vs Stands

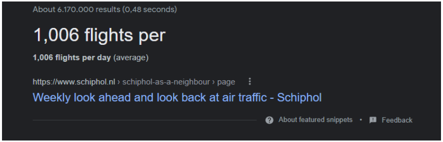


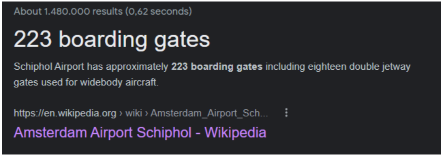
Credits: *The recoverable robust stand allocation problem: a GRU airport case study*

The **Stand Allocation Problem (SAP)** is an alternative approach, more suitable when a gate can allocate multiple aircraft (**gate $\neq$ stand**)

Playing with numbers

Let us focus on **AMS**





Let us assume 500 aircraft $\rightarrow |\mathcal{A}| = 500$ and that each aircraft can only have access to one-fourth of the gates $\rightarrow |\mathcal{G}_i| = 50$

Decision variables $x \sim |\mathcal{A}| \times |\mathcal{G}_i| = 500 \times 50 = 25,000$

Decision variables $y \sim |\mathcal{A}| \times |\mathcal{A} - 1|/2 \times |\mathcal{G}_i| \times |\mathcal{G}_j| \sim 300,000,000$

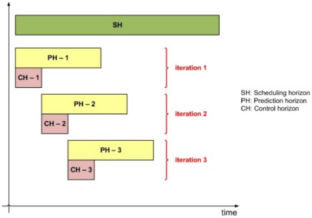
***: $|\mathcal{A}| \times |\mathcal{A} - 1|/2$ as y_{a_1,g_1,a_2,g_2} is equal to y_{a_2,g_2,a_1,g_1} and hence we can only define the first decision variable**

If we were to use flights or activities, the size of the problem might grow even further

GAP: Solution methods

- exact methods: **Branch and Bound**
- approximate methods: **meta-heuristics**
- combinations of the two: **math-heuristics**

Dividing the planning horizon (e.g., one day) into smaller portions: **"Divide and conquer approach": rolling horizon solution method**



Credits: *scienceDirect*

The rolling horizon approach is also beneficial to use **updated information (such as delayed flights) when re-planning**

Recap

- We introduced the Gate Assignment Problem (GAP)
- We assessed that it is a complex problem, with different stakeholders and contrasting interests
- We studied a basic MILP formulation
- We assessed situations where such formulation might be limiting
- We linked the GAP to the Stand Allocation Problem (SAP)
- We wrapped up with some practical considerations

AE4446: *Airport and Cargo Operations* - Gate Assignment

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https://blog.klm.com/nl/10-dingen-die-je-niet-wist-over-schiphol/